

RESEARCH PROJECT

Battery and Hydrogen-Based Storage for Renewables Integration

Characterization of aging
and monitoring of storage systems

Alexandre Ravey · Oumaya Baala · Élodie Pahon

Hybrid storage adapted to renewables

Combining lithium-ion batteries and water electrolyser to recover Intermittent electricity, with safe, efficient and sustainable management

Batteries

Alexandre Ravey

Track temperature, pressure and swelling of the cells

Identify the conditions operating criticisms

Learning

Oumaya Baala

Predicting degradation and detect anomalies

Adapt cycles and operating points

PEM Electrolyzer

Élodie Pahon

Characterizing aging through accelerated testing

Modeling degradation and improve sustainability

Common supervision to preserve the life of the two subsystems

Project resources and roadmap

EXPERIMENTAL AND NUMERICAL MEANS

Battery Instrumentation

Electrical, thermal, and mechanical measurements
Embedded module and communicating BMS

PEM electrolyzer bench

3 kW pressurized demonstrator
Accelerated Aging Protocols

Acquisition and processing

Centralized data, predictive models
and learning control strategies

PROVISIONAL TIMETABLE · 3 YEARS

■ Year 1

Instrumentation of the two benches
First battery characterizations

■ Year 2

Electrolyzer Tests; Module
with temperature and pressure sensors

■ Year 3

Models, communicating BMS
and demonstrator validation

Tutored Project Proposal

GENERAL CONTEXT

The BAT-HYBRID project aims to design a battery management system BMS—an advanced supervision and management unit—for a system comprising an intermittent renewable energy source coupled with an electrolyzer and a storage component consisting of lithium-ion batteries

The project is divided into three main parts:

- a) a battery component
- b) a water electrolysis component
- c) *a supervision and control strategy component based on artificial intelligence capabilities for enhanced durability*

Tutored Project Proposal

DEVELOPMENT OF THE INTELLIGENT MONITORING TOOL

The monitoring tool will manage the collection of measurements and their transmission to a centralized processing system:

- The transmitted data will be processed using data science algorithms
- This data will then be analysed using artificial intelligence models
- The ultimate objective is to develop a system capable of predicting changes in cell conditions, detecting potential failures at an early stage, and dynamically optimizing charge and discharge cycles

This links the system to the intermittent behavior of the electrical source being stored. The electrolyzer's operating phases will also be defined based on the recommendations of the control and monitoring algorithm

Tutored Project Proposal

M1 TUTORED PROJECT TIMELINE

■ Year 1 (M1 students)

■ Fall Semester

- Join the team working on batteries
- Collaborate with the Master's student in Energy on data collection from the test bench

■ Spring Semester

- Understand the characteristics of battery cell data
- Test data fusion and processing algorithms, first using synthetic data, then using measurement data if available
- Propose an initial model for predicting battery cell state

■ Year 2 (M2 students)

■ Fall Semester

- Derive an enhanced version of the proposed battery cell state prediction model including the detection of potential failures, and the optimisation charge and discharge cycles

■ Spring Semester (**Master Internship**)

- Propose a monitoring and control tool to predict battery cell state, detect failures at early stage, and optimise charge and discharge cycles
- Validate the model on real testbed
- Draft a scientific paper to be submitted to an appropriate conference

Do you have any questions?

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Coating design for porous modular structures: improving shape fidelity

A programmable matter is a matter that can be programmed to change its physical properties. It allows the creation of tangible shape-shifting objects that can be reshaped at will and change their functionality to accommodate different tasks or to adapt to their environment. Building large structures efficiently often relies on porous scaffolding: a coarse structure formed from regular meta-modules arranged in a regular lattice, which allows the concurrent flow of modules inside it and reduces module travel distances during reconfiguration. This efficiency comes at a cost: because the scaffold is built from meta-modules rather than individual modules, its outer surface only approximates the target shape at the coarse resolution of the meta-module, with the discretization error appearing as jagged edges, missing overhangs, or over-filled corners along the boundary.

The objective of this project is to design a **coating**: a layer of different meta-modules wrapped around the porous core structure to correct this discretization error and better approximate the target shape at higher resolution [1], [2], [3]. The idea is to keep the coarse porous core intact and only refine its outer boundary. This requires finding a meta-module (or family of meta-modules) whose shape can be locally adapted while remaining compatible with the underlying lattice, structurally attachable to the core, and buildable without blocking the tunnels the core was designed to preserve. The proposed methods will be implemented and evaluated on VisibleSim: an event-based simulator for modular robots.

Requirements:

- Good English reading and writing skills
- Good knowledge of C++ programming
- Knowledge of geometric or combinatorial optimization methods is a plus

References:

- [1] P. Thalamy, B. Piranda, and J. Bourgeois, “3D coating self-assembly for modular robotic scaffolds,” *IEEE Int. Conf. Intell. Robots Syst.*, pp. 11688–11695, 2020, doi: 10.1109/IROS45743.2020.9341324.
- [2] P. Thalamy, B. Piranda, and J. Bourgeois, “Engineering efficient and massively parallel 3D self-reconfiguration using sandboxing, scaffolding and coating,” *Robot. Auton. Syst.*, vol. 146, p. 103875, 2021, doi: 10.1016/j.robot.2021.103875.
- [3] J. Bassil, B. Piranda, A. Makhoul, and J. Bourgeois, “A new porous structure for modular robots,” in *Proceedings of the 21st International Conference on Autonomous Agents and Multiagent Systems*, 2022, pp. 1539–1541.

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Multimodal Digital Twin for Distributed Modular Robots

Research Proposal (M2 - IoT)

Context

Distributed Modular Robotic Systems (DMRS) are made of autonomous modules that assemble, disassemble and reconfigure to change shape and behavior [1]. Blinky Blocks is one such platform, where each module only communicates with its direct neighbors [2]. As a result, a module knows its own state and local connections, but has no way of knowing, for instance, that an obstacle sits next to the structure or that there isn't enough room to reconfigure. This limits monitoring, fault detection and reconfiguration planning in a complex environment.

In earlier work [3] we built a digital twin for Blinky Blocks that reconstructs the physical assembly in real time, using a physical layer, an event-based communication layer, and a virtual representation in VisibleSim. This project builds on that foundation.

Objective

The goal is to extend the digital twin so it captures more than just the current configuration, combining information from multiple sources and adding a first layer of AI-based analysis.

1. Multimodal perception. Combine what the modules already report with external data, mainly from a camera, to get a more complete and reliable picture of the robot and its environment. This makes it possible to cross-check the configuration reconstructed from module communications against what the camera observes, and to capture information no module could know on its own, such as a nearby obstacle. The main question here is how to fuse these sources within the digital twin.

2. AI-based anomaly detection. Use the history of configurations, events and sensor data already collected by the twin to detect unusual behavior. For example, a model that learns what a normal structure looks like and flags a configuration that doesn't fit, such as a loose connection or a failed module.

Requirements

This is a research project for an M2 student in IoT.

- **Required:** C/C++ and Python, background in distributed systems, comfortable with VisibleSim, basic understanding of embedded sensors, good written and spoken English, and strong communication and writing skills (for reporting and documenting results).
- **A plus:** exposure to machine learning or image processing (OpenCV).
- **Qualities:** autonomous, motivated, and curious.

Contact

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References

- [1] S. C. Goldstein, J. D. Campbell, and T. C. Mowry. Programmable matter. *IEEE Computer*, 38(6):99–101, June 2005.
- [2] B. T. Kirby, M. Ashley-Rollman, and S. C. Goldstein. Blinky Blocks: a physical ensemble programming platform. In *CHI'11 Extended Abstracts on Human Factors in Computing Systems*, pp. 1111–1116. ACM, 2011.
- [3] P. Tannoury et al. Development of a digital twin for Blinky Blocks. 2026 (in preparation).

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References

- [1] S. C. Goldstein, J. D. Campbell, and T. C. Mowry. Programmable matter. *IEEE Computer*, 38(6):99–101, June 2005.
- [2] B. T. Kirby, M. Ashley-Rollman, and S. C. Goldstein. Blinky Blocks: a physical ensemble programming platform. In *CHI'11 Extended Abstracts on Human Factors in Computing Systems*, pp. 1111–1116. ACM, 2011.
- [3] P. Tannoury et al. Development of a digital twin for Blinky Blocks. 2026 (in preparation).

Matter allocation in porous modular robotic structures

A programmable matter is a matter that can be programmed to change its physical properties [1]. It allows the creation of tangible shape-shifting objects that can be reshaped at will and change their functionality to accommodate different tasks or to adapt to their environment. The main challenge for self-reconfigurable modular robot-based programmable matter is self-reconfiguration. It consists of finding the sequences of motions that each robot will execute to transform an initial configuration into a given goal configuration

The difficulty of the self-reconfiguration problem lies in the properties of a limited memory space and the locality of the information, which requires modules to communicate to cooperate and coordinate their motions. One possible self-reconfiguration scheme consists of building objects using a porous scaffolding structure formed using regular meta-modules placed in regular lattices, which allows the concurrent flow of modules inside of it and the reduction of the travel distances of modules [2], [3], [4].

Porous structures allow the storage of matter (modules) inside the empty free volumes of the structure. The objective of this project is to study and propose matter allocation methods to optimize the filling space. For instance, filling positions can be chosen to ensure structural stability by adding weight to the base of the structure. Alternatively, allocation strategies can be designed to facilitate reconfiguration, or maximize accessibility of stored modules. The methods developed should account for geometric and blocking constraints, load distribution, and potential reconfiguration scenarios. The proposed methods will be implemented on VisibleSim: an event-based simulator for modular robots.

Requirements:

- Good english reading and writing skills
- Good knowledge C++ programming
- Knowledge in resource allocation methods is a plus

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References:

- [1] J. Bourgeois *et al.*, “Programmable matter as a cyber-physical conjugation,” in *2016 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, IEEE, Oct. 2016, pp. 002942–002947. doi: 10.1109/SMC.2016.7844687.
- [2] J. Bassil, B. Piranda, A. Makhoul, and J. Bourgeois, “ASAPs: asynchronous hybrid self-reconfiguration algorithm for porous modular robotic structures,” *Auton. Robots*, vol. 48, no. 7, p. 16, 2024.
- [3] J. Bassil, B. Piranda, A. Makhoul, and J. Bourgeois, “RePoSt: Distributed Self-Reconfiguration Algorithm for Modular Robots Based on Porous Structure,” in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, Oct. 2022, pp. 12651–12658. doi: 10.1109/IROS47612.2022.9981212.
- [4] P. Thalamy, B. Piranda, and J. Bourgeois, “Engineering efficient and massively parallel 3D self-reconfiguration using sandboxing, scaffolding and coating,” *Robot. Auton. Syst.*, vol. 146, p. 103875, 2021, doi: 10.1016/j.robot.2021.103875.

Neural Cellular Automata For Modular Robots

Lattice-based modular robots (Catoms3D, Datoms, Blinky Blocks) are built from thousands of identical units that can only sense and talk to their immediate neighbors. Many useful facts about the ensemble's structure are defined globally: which part of the overall shape a module belongs to, which modules are load-bearing, which are articulation points. These are expensive to compute exactly from local communication alone, and the computation degrades badly when modules fail silently.

A recent paper by Moreno et al. [1] showed that a Neural Cellular Automaton [2] (a single, shared local update rule applied repeatedly across the lattice) lets several hundred physical cubic modules classify their own overall shape and detect missing neighbors using only six-face local communication. No global positioning is needed, and the approach tolerates module failure well.

Piranda et al. [3] already mapped a small feed-forward ANN as a distributed, agent-based computation directly onto real BlinkyBlock hardware [4] [5], letting modules recognise a lit shape (square, disk, or triangle) using only local neighbor-to-neighbor messages, and matched a centralised network's output exactly. This shows the same core mechanism already works on our own platform, not just in simulation.

This project asks how far that same idea (purely local, learned rules) can be pushed as a standalone inference capability. The target is per-module decision-making, not a single collapsed shape label:

1. Part membership. Given a fixed structure, can each module learn which functional part of the overall shape it belongs to (leg, surface, support), using only what it and its near neighbours can observe? Ground-truth labels come from existing 3D shape datasets, such as ShapeNet's part annotations.
2. Structural role detection. Can a module learn to detect whether it is load-bearing, or whether it is an articulation point whose failure would disconnect the ensemble?
3. Damage and anomaly detection. Can a module detect, from local signal alone, that a neighbour has failed silently, and localise the direction of the gap? This would reproduce the reference paper's damage-detection result, but evaluated systematically rather than anecdotally.

Requirements:

- Good english reading and writing skills
- Good knowledge of Python and Deep Learning Libraries
- Willingness to work with C++ (VisibleSim)
- Knowledge of neural networks and distributed algorithms is a plus

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References:

- [1] R. Moreno, A. Faíña, S. Sudhakaran, K. Walker, and S. Risi, "Smart cellular bricks for decentralized shape classification and damage recovery," *Nat. Commun.*, vol. 17, no. 1, p. 5932, Jul. 2026, doi: 10.1038/s41467-026-75166-7.

- [2] E. Randazzo, A. Mordvintsev, E. Niklasson, M. Levin, and S. Greydanus, "Self-classifying MNIST Digits," *Distill*, vol. 5, no. 8, p. e00027.002, Aug. 2020, doi: 10.23915/distill.00027.002.
- [3] B. Piranda, M. A. Nemer, A. Makhoul, and J. Bourgeois, "AI4PM: Distributed and Intelligent Programmable Matter," in *Advanced Information Networking and Applications*, vol. 246, L. Barolli, Ed., in Lecture Notes on Data Engineering and Communications Technologies, vol. 246. , Cham: Springer Nature Switzerland, 2025, pp. 116–127. doi: 10.1007/978-3-031-87766-7_11.
- [4] B. Kirby, M. Ashley-Rollman, and S. C. Goldstein, "Blinky Blocks: A physical ensemble programming platform," *Conf. Hum. Factors Comput. Syst. - Proc.*, pp. 1111–1116, 2011, doi: 10.1145/1979742.1979712.
- [5] F. Lassabe, D. Dhoutaut, B. Piranda, O. Kouchnarenko, and J. Bourgeois, "Building a First Prototype of a Multi-scale Modular Distributed Display," IEEE Congress on Cybermatics, IEEE Computer Society, Aug. 2024, pp. 276–281. doi: 10.1109/iThings-GreenCom-CPSCoM-SmartData-Cybermatics62450.2024.00064.

Automatic generation of motion operations for generated meta-module designs

Self-reconfiguration in lattice-based modular robots is often organized around meta-modules: small, fixed clusters of modules that move and reassemble collectively rather than module by module. Designing a meta-module has two separate parts: its shape (the geometric arrangement of modules) and its operation (the sequence of individual module motions that lets it actually move, reconfigure, or propagate motion through a lattice) [1], [2], [3],[4].

Recent work in the group has produced a formal framework and automated methods (genetic algorithms) for generating diverse, valid meta-module shapes, including porous designs that let motion tunnel through a larger structure.

The objective of this project is to automate this second step: given an already-generated meta-module shape as input, automatically derive valid operations for it. This means searching over sequences of elementary module moves that are individually legal, that keep the meta-module connected throughout, and that optimize for a target behaviour such as net displacement, flowtime and number of motions. The methods developed should account for the module's motion constraints and the risk of disconnection mid-sequence [5]. The proposed methods will be implemented and evaluated on VisibleSim [6]: an event-based simulator for modular robots, using the existing formal meta-module representation as the interface between design and operation search.

Requirements:

- Good English reading and writing skills
- Good knowledge of C++ programming
- Knowledge of search or optimization methods and advanced algorithms

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References:

- [1] P. Thalamy, B. Piranda, and J. Bourgeois, "Engineering efficient and massively parallel 3D self-reconfiguration using sandboxing, scaffolding and coating," *Robot. Auton. Syst.*, vol. 146, p. 103875, 2021, doi: 10.1016/j.robot.2021.103875.
- [2] J. Bassil, B. Piranda, A. Makhoul, and J. Bourgeois, "ASAPs: asynchronous hybrid self-reconfiguration algorithm for porous modular robotic structures," *Auton. Robots*, vol. 48, no. 7, p. 16, 2024.
- [3] J. Bassil, B. Piranda, A. Makhoul, and J. Bourgeois, "RePoSt: Distributed Self-Reconfiguration Algorithm for Modular Robots Based on Porous Structure," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, Oct. 2022, pp. 12651–12658. doi: 10.1109/IROS47612.2022.9981212.
- [4] G. A. Ahad, N. Awad, J. Bourgeois, J. Werfel, and B. Piranda, "Learning Fault-Tolerant Navigation with Self-Reconfiguring Modular Robots," in *Advanced Information Networking and Applications*, vol. 295, L. Barolli, W. K. G. Seah, and I. Woungang, Eds., in *Lecture Notes on Data Engineering and Communications Technologies*, vol. 295, Cham: Springer Nature Switzerland, 2026, pp. 99–109. doi: 10.1007/978-3-032-23271-7_9.
- [5] B. Piranda and J. Bourgeois, "Designing a quasi-spherical module for a huge modular robot to create programmable matter," *Auton. Robots*, vol. 42, no. 8, pp. 1619–1633, Dec. 2018, doi: 10.1007/s10514-018-9710-0.
- [6] P. Thalamy, B. Piranda, A. Naz, and J. Bourgeois, "VisibleSim: A behavioral simulation framework for lattice modular robots," *Robot. Auton. Syst.*, vol. 147, p. 103913, Jan. 2022, doi: 10.1016/j.robot.2021.103913.

Privacy-Preserving Semantic Knowledge Transfer for Heterogeneous Agentic AI in IIoT

Research Project Proposal & Literature Review Baseline

Supervisor:	Nancy AWAD	Date:	September 2026
Domain:	Agentic AI, Federated Learning (FL), Industrial Internet of Things (IIoT)	Target Output:	Academic Paper / Research Presentation Framework
Core Focus:	Federation over Text, Semantic Distillation, Heterogeneous LLM Collaboration		

1. Executive Summary & Context

Modern Industrial Internet of Things (IIoT) environments are transitioning toward autonomous operational management driven by Large Language Model (LLM)-based **AI Agents**. These edge agents autonomously perform multi-step planning, real-time fault diagnosis, code execution, and tool invocations. However, scaling collaborative intelligence across decentralized edge sites faces severe architectural barriers. Traditional Federated Learning (FL) methods (e.g., FedAvg, FedLoRA) depend on averaging model parameter weights, which fails when edge nodes operate heterogeneous LLM architectures and under strict data privacy regulations.

This research introduces **Fed-SemanticIIoT**, a novel framework that shifts knowledge sharing from numerical weight updates to *semantic trajectory distillation*. Edge agents execute local operational trajectories, extract anonymized, natural-language procedural skill patches, and federate these explicit rules across the network to build a shared global memory bank without exposing raw proprietary sensor telemetry.

2. The IIoT Agent Trilemma (Problem Statement)

1. Model Heterogeneity Edge devices deploy distinct models ranging from lightweight Small Language Models (SLMs) to domain-specific LLMs (e.g., Llama-3 8B vs. Qwen-2.5 7B). Standard weight-averaging (FedAvg) is mathematically impossible across different architectures.	2. Proprietary Data Privacy Industrial tenants cannot transmit raw terminal traces, internal network topologies, or sensor telemetry to a central server due to strict trade secret policies. Weight updates also remain vulnerable to inversion attacks.	3. Environmental Non-IID Drift Local operating conditions vary dramatically across plant floors. Agents isolated in local silos repeatedly waste compute rediscovering solutions to zero-day anomaly patterns that peer nodes have already resolved.
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Research Objective: Design a model-agnostic, privacy-preserving knowledge exchange protocol enabling decentralized AI agents to share abstract operational strategies across heterogeneous IIoT nodes, guaranteeing zero exposure of proprietary underlying data while drastically reducing network communication overhead.

3. Key Research Questions (RQs)

- RQ1 (Cross-Model Skill Transfer):** How can natural language skill distillation enable cross-architecture knowledge sharing between heterogeneous IIoT agents without parameter alignment?
- RQ2 (Privacy & Context Sanitization):** What automated sanitization protocols ensure that distilled trajectory patches strip out 100% of proprietary operational data (e.g., IP addresses, device IDs, specific parameter bounds)?

- **RQ3 (Conflict Resolution & Aggregation):** How can a federated aggregator detect, reconcile, or reject conflicting or hallucinated skill updates generated by agents in diverse local factory environments?

4. Proposed System Architecture & Workflow

1. **Local Trajectory Execution:** Edge agents execute multi-step operational tasks using local tools (*Observe → Reason → Call Tool → Correct*).
2. **Semantic Skill Distillation:** Upon solving a novel operational error, a local distiller module extracts a generalized procedural insight (e.g., *"If temperature spikes during phase shift X, adjust valve regulator Y before resetting error flag Z"*).
3. **Context Scrubbing:** A guardrail filter strips site-specific entity variables, network credentials, and proprietary thresholds.
4. **Federated Memory Alignment:** A central/peer aggregator merges valid skill patches into a **Global Heuristic Library**, which edge agents retrieve at inference time as context-augmented memory.

5. Methodological Evaluation Plan

- **Simulation Setup:** Multi-agent node simulation using Python and Ollama running lightweight open-source models (Llama-3-8B, Qwen-2.5-7B, Mistral-7B).
- **Benchmarks:** Industrial fault-detection benchmarks (e.g., TON_IoT, IoT-23) and synthetic tool-use execution environments.
- **Metrics:** Task Success Rate (%), Convergence Speed (steps to resolution), Privacy Leakage Score (context reconstruction rate), and Communication Bandwidth Consumption (KB transferred vs. MB/GB in FedLoRA).

6. Key Academic References & Literature Base

- Agent Trajectories** S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, *"ReAct: Synergizing Reasoning and Acting in Language Models,"* in Proc. Int. Conf. Learn. Represent. (ICLR), 2023.
- Federation over Text** R. Wang et al., *"Federated In-Context Learning: Iterative Refinement for Improved Answer Quality,"* in Proc. Int. Conf. Machine Learning (ICML), 2025.
- FL & Agents** Z. Zhang et al., *"Federated In-Context LLM Agent Learning,"* arXiv preprint arXiv:2412.08054, 2024.
- LoRA / Parameter FL** L. Yi, H. Yu, G. Wang, and X. Liu, *"FedLoRA: Model-Heterogeneous Personalized Federated Learning with LoRA Tuning,"* arXiv preprint arXiv:2310.13283, 2023.



Master IoT
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Sophia project: Education in critical analysis of SOPHisms using AI

In the age of the Internet and social media, sources of information have multiplied, offering greater accessibility to data and public forums for debate. This development has been accompanied by harmful phenomena, with the emergence of disinformation practices that are both organized, intensified, and automated. This disinformation biases the democratic virtues of these alternative spaces for information and exchange, through the fabrication of fake news on the one hand, and through fallacious argumentation techniques that exploit such information on the other.

Most initiatives focus on education in critical analysis of sources and cross-referencing them. SophIA project aims to immunize minds against fallacious argumentation techniques. We therefore propose in this project to conceive and develop a playful mobile/web application using AI chatbots, where players are confronted with a series of argumentative constructs and then invited to expose the fallacious techniques used.

Requested skills

- Web and/or mobile development
- Containerization and networking knowledges

Hakim MABED

Tutored Project Topics

Multimodal Mobility, Data Science, Optimization and Intelligent Transportation

Project 1 — Data Visualization of Mobility Services in Cross-Border Areas

France–Switzerland, International Airports and Multimodal Mobility

Target students:

- 1 or 2 M1 students, or
- 1 or 2 M2 students, or
- a mixed team of 1 M1 + 1 M2 student

Keywords: Data Science, Big Data, IoT, Data Visualization, Web APIs, Multimodal Transportation, Mobility as a Service

Supervision: Philippe Canalda, Ahmed-Imad Hamaili (Mu-CAR.i Startup), Hakim Mabed

Context

Within the framework of the **Mu-CAR (2024–2027)**, **SATURN (2025–2028)** and **E-TAKHAWAL-MaaS (2026–2029)** projects, as well as the **SERM project currently being submitted with the Metropolitan Cluster (2026–2032)**, the aggregation of heterogeneous and dynamic multimodal transportation data is a key challenge.

Such data may include **bus, regional and high-speed trains (TER, TGV), demand-responsive transport (DRT), bike sharing, car sharing, carpooling, walking and mobility services for people with reduced mobility (PRM)**. Efficient aggregation of these heterogeneous data sources is a fundamental prerequisite for any comprehensive mobility service.

The **Mu-CAR_MaaS platform**, developed within the CNRS FEMTO-ST/DISC/OMNI laboratory, is currently progressing towards **TRL 5**, corresponding to a validated laboratory prototype ready for real-world experimentation or an initial operational deployment.

Experiments are planned in **Dakar between January and June 2028**, and potentially within the **Belfort–Montbéliard–Héricourt–Delle metropolitan area** from the end of 2027.

Project objectives

The project aims to develop a **static and dynamic aggregation of French and Swiss mobility data** across the Jura Arc and, subsequently, along the entire French–Swiss border, including the cantons of **Basel and Lausanne**.

The project will consider:

- public and private parking facilities;
- potential pickup and drop-off locations;
- multimodal transfer points;
- opportunities for modal shift;
- French and Swiss public transportation;
- national and international airport data, with **EuroAirport Basel–Mulhouse** as the primary prototype case.

The project team already has access, through Web APIs, to Swiss transportation supply data, excluding individual usage data for obvious GDPR and privacy reasons.

The objective is to provide a **mobility visualization system** covering the Jura Arc and the French–Swiss cross-border region, while extending the Mu-CAR_MaaS platform to support two major types of queries.

Use case 1 — Airport-related multimodal mobility

When a user selects a domestic or international flight arriving at or departing from Basel–Mulhouse Airport, the system should propose the corresponding multimodal mobility request and multimodal access routes, following the Mu-CAR approach:

(src) Mode_1 Stop_1 [h–, h+] NbPU/NbD → [(via) Mode_i Stop_i [h–, h+] NbPU/NbD →]* (dst) Mode_N Stop_N [h–, h+] NbPU/NbD

Use case 2 — Cross-border multimodal journeys

When a user selects a transport stop in Switzerland, the system should propose:

- an outbound multimodal itinerary;
- a return multimodal itinerary;
- potential carpooling opportunities before reaching the border;
- potential shared-mobility opportunities at or around the border.

Benchmarking

A benchmarking study will be conducted to identify:

- academic and commercial competitors;
- their main functionalities;
- their limitations;
- estimated solution costs;
- potential business and economic models.

Data aggregation will initially be implemented in a **static mode**, before moving towards **dynamic aggregation**.

Mu-CAR platform integration

In parallel, students will discover and use the Mu-CAR platform. This will facilitate the integration of Mu-CAR_MaaS into the benchmarking study and help students identify automatable processes and determine the appropriate data aggregation formats.

Particular attention will be paid to existing standards and normalized formats, including:

- GTFS
- GBFS
- GTFS-RT
- SIRI
- NeTEx

Expected deliverables

- A research or technical report continuously updated throughout the project.
- A Git repository containing all developments and tests.
- At least one scientific and/or technical paper submitted to a conference ranked at least B or C.
- Participation in 1 to 3 scientific outreach events, such as Festival Inouïh, La Science en Fête, Researchers' Night, EDEN.i events, Student IEEE Chapter events, etc.

Further opportunities: Possibility of continuing the work through **one or two internships**.

Project 2 — Tabu Search Mechanisms for Multimodal Carpooling Algorithms

Extension of a Greedy-Based Flexible Carpooling Algorithm with Exact Insertion

Target students:

- 1 M1 student, or
- 1 M2 student

Keywords: Operations Research, Combinatorial Algorithms, Exact and Approximate On-Demand Transportation Optimization, Intelligent Transportation Systems, Data Science, Big Data, IoT, Web APIs

Supervision: Philippe Canalda, PhD student within the E-TAKHAWAL_MaaS Senegalese project (October 2026–December 2029), Frédéric Gaspoz (observer, University of Sion, Switzerland), Oumaya Baala, Hakim Mabed

Context

Within the framework of the **Mu-CAR (2024–2027)** and **E-TAKHAWAL-MaaS (2026–2029)** projects, as well as the **SERM project currently being submitted with the Metropolitan Cluster (2026–2032)**, a flexible and dynamic carpooling service integrated with a global transportation offer is being prototyped.

A multimodal travel information service is also being developed within the **Région-Eiphi project (2024–2027)**.

These services are expected to be experimentally evaluated in **Dakar between January and June 2028** and potentially deployed operationally within the **Belfort–Montbéliard–Héricourt–Delle metropolitan area** from the end of 2027.

The Mu-CAR_MaaS platform, developed within the CNRS FEMTO-ST/DISC/OMNI laboratory, is progressing towards **TRL 5**.

Existing algorithmic component

The Mu-CAR_MaaS platform already includes the **Mu-CAR_GbFCA Web API**, implementing a:

Greedy-Based Flexible Carpooling Algorithm with Exact Insertion using a Cut-and-PriceShare Dynamic Programming Algorithm.

The algorithm solves, approximately and in linear complexity, dynamic carpooling problems involving:

- 0-intramodal transportation;
- 0-to-2 multimodal transportation.

The component currently has a maturity level above **TRL 3**, with a complete test suite allowing functional testing as well as performance benchmarking.

The solution is implemented in **Python**, while integration and usage involve **JavaScript, Python and PHP**.

Research problem

The number of operational solutions currently generated by the system is limited.

The objective of this project is therefore to introduce a **Tabu Search mechanism**, based on tabu requests and/or tabu carpooling crews, in order to prevent the system from repeatedly proposing the same matching and the same generated itinerary.

Beyond the Tabu mechanism itself, the project aims to generate **diversified, operational and relevant mobility solutions** for users of a complete and innovative MaaS service.

Main objectives

The project will involve:

1. Designing and formalizing a Tabu Search mechanism suitable for the existing carpooling algorithms.
2. Integrating the mechanism into the Mu-CAR_GbFCA architecture.
3. Evaluating the diversity and quality of generated solutions.
4. Measuring computational performance.
5. Comparing the new approach with the existing algorithm.
6. Extending the current test suite to cover the newly introduced functionalities.

Benchmarking

A benchmarking study will identify relevant:

- academic approaches;
- potentially commercial solutions;
- metaheuristic and combinatorial optimization techniques;
- approaches for generating diversified solutions in transportation optimization.

Development strategy

A first prototype will be developed and evaluated in **Python**.

Although a complete and minimal test set already exists for flexible carpooling, it will need to be extended to properly validate the new functionalities.

Students will also become familiar with the Mu-CAR_MaaS platform and its integration into benchmarking workflows.

In a second stage, the algorithm may be reimplemented in a more operational programming language such as **C or C++**, with the possibility of subsequently investigating **real-time execution**.

Expected deliverables

- A research or technical report continuously updated throughout the project.
- A Git repository containing all developments and tests.
- At least one scientific and/or technical paper submitted to a conference ranked at least B or C.
- At least one algorithm, including:
 - formal specification;
 - operational implementation;
 - functional evaluation;
 - qualitative evaluation;
 - performance evaluation;
- test datasets;
- analysis of experimental results.
- Participation in 1 to 3 scientific outreach events, such as Festival Inouïh, La Science en Fête, Researchers' Night, EDEN.i events, Student IEEE Chapter events, etc.

Further opportunities: Possibility of continuing through **one or two internships**.

PhD opportunity: Potential continuation towards a **PhD thesis**.

Project 3 — Multi-Objective Optimization of Trip Sharing and Carbon Emission Reduction

Application to Mission Management Services within Université Marie et Louis Pasteur (UMLP)

Target students:

- Preferably 1, 2 or 3 M1 students

Keywords: Data Science, Big Data, IoT, Anonymization, Web APIs, Multi-Criteria Optimization, Decarbonization, Mobility, Carpooling

Supervision: Philippe Canalda, Vincent Bertrand (CRESE), Eugen Dedu, Anna Joliot (FEMTO-ST PhD student), Julien Montillot (Vice-President, UMLP)

Context

Within the framework of the **REEL.i Chair** and the **EDEN.i network**, research activities addressing transport decarbonization are rapidly expanding.

The **Mu-CAR (2024–2027)** and **SATURN (2025–2028)** projects contribute directly to these efforts.

Through these projects, the research team has access to the **Mu-CAR_GbFCA Web API**, which provides dynamic flexible carpooling for:

- 0-intramodal transportation;
- 0-to-2 multimodal transportation.

More specifically, the service can create carpooling crews and generate flexible itineraries while managing:

- changes in driver/passenger roles;
- designated or non-designated drivers;
- pickup and drop-off constraints;
- authorized detours;
- distance constraints;
- travel-time constraints;
- vehicle capacity;
- dynamic or non-dynamic operation.

This flexible carpooling approach is particularly suitable when a driver is designated—or needs to be designated—to share their vehicle while respecting **dynamic time windows (DTWs)** and vehicle-capacity constraints.

The proposed journeys are generally **seamless**, meaning that passengers and drivers remain in the same vehicle throughout the carpooling segment. Carpooling may nevertheless be connected to another transportation mode for the first or last mile.

Application within UMLP

Within **Université Marie et Louis Pasteur (UMLP)**, several services, teaching departments and the STGi faculty currently use a dematerialized mission-management system.

The current carpooling mechanism essentially identifies whether mission destinations exactly coincide, based on criteria such as:

- same date;
- same destination city.

A message is then sent to potential members of a carpooling crew shortly before the mission.

The **DOM system** manages mission requests and subsequent reimbursement requests once the mission has been authorized.

Several years of historical usage data are available.

Project objectives

The project will investigate how multi-objective optimization can be used to maximize:

- trip sharing;
- transport decarbonization;
- reduction of mission-related costs;
- social interaction between university populations;
- overall efficiency of mobility organization.

Task 1 — Simulation and synthetic mobility datasets

The project team will develop simulations of:

- university staff;
- home locations;
- workplace locations;
- travel events;
- mission requests;
- different population profiles.

Synthetic and parameterizable datasets will be generated to test different carpooling configurations and itinerary-generation strategies.

This will make it possible to estimate carpooling rates for different community profiles, such as:

- researchers;
- technical staff;
- administrative staff;
- students;
- invited speakers and external participants.

Task 2 — Extraction and anonymization of DOM data

Several years of DOM data will be extracted and anonymized.

The objective will be to study different **transport decarbonization policies** and evaluate different strategies for organizing carpooling.

The project will pay particular attention to privacy, anonymization and responsible use of mobility data.

Task 3 — Integration into the DOM system

The project will investigate the integration of the Mu-CAR optimization service into DOM through:

- **Mu-CAR_GbFCA Web API;**
- **Mu-CAR showroom/dashboard Web APIs.**

The team will investigate the integration of exact carpooling solutions for:

- unimodal carpooling;
- 1-intramodal carpooling;
- carpooling involving a parking-based transfer or modal shift.

These recent research developments will be tested in connection with the work conducted by **Anna Joliot**.

Behavioral intervention and nudging

The Web-service calls will be complemented by messages and **behavioral nudges** designed to evaluate factors influencing mobility behavior.

The ultimate objective is to investigate whether mobility recommendations can encourage behavioral changes supporting:

- climate-change mitigation;
- carbon-emission reduction;
- reduction of mission-related expenses;
- increased social interaction and community building.

Expected deliverables

- A research or technical report continuously updated throughout the project.
- A Git repository containing all developments and tests.
- At least one scientific and/or technical paper submitted to a conference ranked at least B or C.
- At least one algorithm, including:
 - formal specification;
 - functional, qualitative and performance evaluation;
 - test datasets;
 - analysis of experimental results.
- Participation in 1 to 3 scientific outreach events, such as Festival Inouïh, La Science en Fête, Researchers' Night, EDEN.i events, Student IEEE Chapter events, etc.

Further opportunities: Possibility of continuing through **one internship**.

PhD opportunity: Potential continuation towards a PhD in **combinatorial optimization, service design and service evaluation**.

Entrepreneurial opportunity: Potential for an **entrepreneurial continuation**.

Project 4 — Machine Learning, Data Visualization, Mobility Metrics and Performance Analysis

Application to Multimodal Mobility Data and Usage in Fragmented Territories

Target students:

- Preferably 1 M2 student
- 1 M1 student may also be considered

Keywords: Combinatorial Algorithms, Heuristics, Data Science, Big Data, Metrics, IoT, Intelligent Transportation Systems, Multimodality, Real-Time Systems, Time Constraints, Machine Learning

Supervision: Anna Joliot (FEMTO-ST PhD student), Philippe Canalda, Franco-German / SMYLE Consortium (V. Kaufmann, M. Bierlaire, etc.), Hakim Mabed

Context

The PhD thesis of **Anna Joliot** is entering its third and final year.

Her research, at the intersection of applied mathematics and computer science, focuses on **exact MILP-based approaches for carpooling optimization**, including:

1. flexible unimodal carpooling;
2. N-intramodal flexible carpooling, with or without modal transfers;
3. real-time multimodal carpooling.

The territories considered in this research are particularly complex.

They are geographically fragmented by:

- departmental boundaries;
- regional boundaries;
- international borders.

They are also fragmented from a mobility-service perspective, involving:

- multiple transport authorities;
- multiple mobility operators;
- multiple transportation modes;
- heterogeneous populations;

- heterogeneous mobility services.

Research objective

To better characterize these complex territories, the project aims to define, calculate and analyze a set of **mobility metrics and variables**.

The expected results should help guide:

- hybrid research strategies;
- economic policy evaluation;
- behavioral policies;
- mobility-service design;
- machine-learning approaches;
- optimization strategies.

Three main research tasks

Task 1 — Analysis of Anna Joliot's algorithms and datasets

The first task will focus on the algorithms developed by Anna Joliot and on the datasets they manipulate or generate.

Students will investigate:

- algorithmic behavior;
- computational performance;
- characteristics of generated solutions;
- relationships between input data and optimization results;
- relevant performance metrics.

Task 2 — Learning mobility data and mobility-service usage

The second task will investigate machine-learning approaches for learning from mobility data and their evolution over time.

Particular attention will be paid to:

- seasonal variations;
- temporal mobility patterns;
- multimodal usage;
- intelligent mobility-service adoption;
- prediction of mobility demand;
- learning from historical mobility behavior.

The results may be used to feed **Mu-CAR multimodal carpooling push services**.

Task 3 — Academic benchmarking of AI approaches

The third task will consist of a primarily academic benchmarking study of **AI-based approaches** capable of addressing the objectives of the project.

The benchmark may cover:

- machine learning;
- predictive models;
- reinforcement learning;
- hybrid AI;
- heuristic and metaheuristic approaches;
- combinations of machine learning and combinatorial optimization.

Expected deliverables

- A research or technical report continuously updated throughout the project.
- A Git repository containing all developments and tests.
- At least one scientific and/or technical paper submitted to a conference ranked at least B or C.
- At least one algorithm, including:
 - formal specification;
 - operational implementation;
 - functional evaluation;
 - qualitative evaluation;
 - performance evaluation;
- test datasets;
- analysis of experimental results.
- Participation in 1 to 3 scientific outreach events, such as Festival Inouïh, La Science en Fête, Researchers' Night, EDEN.i events, Student IEEE Chapter events, etc.

Further opportunities: Possibility of continuing through **one internship**.

PhD opportunity: Potential continuation towards a PhD in **AI and hybrid combinatorial optimization**.

International research opportunities: The project is connected to the emerging **Franco-Swiss SMYLE consortium**, with potential perspectives involving:

- projects focusing on the Jura Arc;
- INTERREG projects;
- European mobility projects;
- potential collaboration with Germany as a first international target.

Common Research Environment

The four projects are part of a common research ecosystem involving **FEMTO-ST/DISC/OMNI**, the Mu-CAR_MaaS platform, and several national and international research projects.

Depending on the selected project, students may work on a combination of:

- multimodal mobility;
- Mobility as a Service (MaaS);
- intelligent transportation systems;
- data science and Big Data;
- Web APIs;

- IoT;
- combinatorial optimization;
- exact and heuristic algorithms;
- machine learning and hybrid AI;
- data visualization;
- mobility metrics;
- transport decarbonization;
- real-time transportation services.

The projects are intended to combine **research, software development, experimentation, benchmarking and scientific dissemination**, with opportunities for continuation through internships, PhD theses and, in some cases, entrepreneurial projects.

Region-Based Routing for Modular Robots

Master 2 IoT Project Proposal

Project Description

Modular robots are composed of several interconnected modules organized into different physical or functional parts. For example, a robot may be divided into Part 1, Part 2, Part 3, etc., where each part contains several communicating modules. A humanoid robot is one possible example, where the robot will be divided into hierarchical regions such as:

Torso → Left Arm → Left Forearm → Left Hand.

The objective of this project is to design and evaluate a region-based routing protocol for modular robotic networks, inspired by geographic and geocast routing protocols, such as GEAR (Geographical and Energy Aware Routing). Instead of sending a message only to a specific node, the protocol will allow messages to be addressed to a region of the robot.

Example route: Part 1 -> Part 2 -> Part 3 -> Target Region

Each module will know its own region and information about its direct neighbors. Packets will be forwarded toward the target region using local information such as region distance, hop count, link quality, network load, and node availability.

The protocol may support:

- **Region Anycast:** deliver the message to one suitable node inside the target region;
- **Region Geocast:** deliver the message to several or all nodes inside the region;
- **Region Unicast:** reach a target region and then a specific node;
- **Fault-aware routing:** select an alternative path when links or modules fail.

The protocol will be implemented and evaluated using VisibleSim, a simulator for modular robotic systems. Several topologies and failure scenarios will be studied. Performance will be evaluated using metrics such as packet delivery ratio, end-to-end latency, number of transmitted packets, routing overhead, and robustness to node or link failures.

For the example of humanoid, A possible routing metric will combine the morphological distance to the destination region with communication parameters such as link quality and network load. Restricted flooding or local dissemination can then be used once the packet reaches the destination region.

For example, a message addressed to LEFT_HAND could follow:

Torso → Left Arm → Left Forearm → Left Hand

without requiring the sender to know the identifiers or exact physical positions of all modules in the hand.

Main Tasks

The students will first study existing geographic and geocast routing approaches for sensor networks, including GEAR and related protocols. A region-based routing model for modular robots will then be designed and implemented. The protocol will be evaluated using the visiblesim simulator.

Different scenarios will be considered, including normal routing, communication with multiple regions, node/link failures, topology changes, and comparison with traditional flooding or shortest-path approaches.

Expected Outcomes

At the end of the project, the students should provide a formal description of the proposed routing protocol, a working prototype or simulation, an experimental evaluation, and a comparison with at least one conventional

routing strategy. An optional extension could investigate dynamic robot reconfiguration, where modules can be added, removed, or moved between regions.

Prerequisites

- Basic knowledge of computer networks and routing protocols;
- Basic knowledge of IoT and Wireless Sensor Networks;
- Programming skills in C/C++ and/or Python;
- Basic understanding of graph algorithms and distributed systems;
- Knowledge of modular robotics or VisibleSim is an advantage but is not mandatory.

The project combines IoT networking, distributed routing, fault tolerance, and modular robotics.

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